P-S-620D
July 9, 1975
SUPERSEDING
Int. Fed. Spec. P-S-00620C
August 10, 1970 and
Fed. Spec. P-S-620B
August 24, 1965
(See 6.3)

FEDERAL SPECIFICATION

SOAP, TOILET (CAKE AND POWDERED)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers cake and powdered toilet soap (see 6.1).

1.2 Classification. Toilet soap shall be of the following types and classes, as specified (see 6.2 and 6.4).

Type I - Cake toilet soap.

Class 1 - White floating cake.
Class 2 - Milled cake.

Type II - Powdered toilet soap.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

L-P-378 - Plastic Sheet and Strip, Thin Gauge, Polyolefin.
FF-D-396 - Dispensers, Soap.
PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-636 - Boxes, Shipping, Fiberboard.

Federal Standards:

Fed. Std. No. 123 - Marking for Shipment (Civil Agencies).
Fed. Test Method Std. No. 536 - Soap and Soap Products (Including Synthetic Detergents): Sampling and Testing.
Fed. Std. No. 595 - Colors.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

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Military Specification:

MIL-L-10547 - Liners, Case, and Sheet, Overwrap, Water-Vaperproof or Waterproof, Flexible.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Laws and Regulations:

21 CFR 1 - Federal Food, Drug, and Cosmetic Act.

(The Code of Federal Regulations (CFR) and the Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, U.S. Goverment Printing Office, Washington, DC 20402. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for issuance thereof.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Standards:

D 460 - Sampling and Chemical Analysis of Soaps and Soap Products.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Finished product. The finished product shall be a nonirritating, thoroughly saponified soap conforming to the physical and chemical requirements of table I, and the other requirements of this specification.

3.1.1 Organic oxidation inhibitor. Type II toilet soap shall contain a minimum of 0.02 percent by weight of an organic oxidation inhibitor. The following inhibitors are acceptable: diphenylamine, orthotolyl-biguanidine, paratertiary amyl phenol, paratertiary butyl phenol, orthodiphenyl biguanidine, butylated hydroxy toluene or monostearic hydrazide. The contractor shall certify that his soap meets this requirement and indicate which inhibitor is being used.

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TABLE I. Physical and chemical requirements

Property	Type I class 1 min.	Type I class 1 max.	Type I class 2 min.	Type I class 2 max.	Type II min.	Type II max.
Moisture and volatile matter; percent by weight	----	24.0	----	15.0	----	6.0
Free alkali (as NaOH), percent by weight	----	0.1	----	0.1	----	0.1
Sodium chloride; percent by weight	----	1.2	----	1.2	----	1.2
Sum of free alkali, total matter insoluble in alcohol, and sodium chloride; percent by weight	----	2.5	----	2.5	----	2.0
Matter insoluble in water, percent by weight	----	0.6	----	1.0	----	0.2
Anhydrous soap, percent by weight	73.5	----	83.0	----	91.0	----
Unsaponified saponifiable matter; percent by weight	----	0.3	----	0.3	----	0.3
Titer, degrees C	34.0	----	36.0	----	36.0	----
Acid number; mg KOH/g mixed fatty acids from soap	215	----	209	----	209	----
Sugar	----	none	----	none	----	none
Rosin	----	none	----	none	----	none
Fineness-weight percent retained on:						
No. 12 US Standard Sieve					----	1.5
No. 40 US Standard Sieve					50.0	----
No. 100 US Standard Sieve					90.0	----

3.2 Form.

3.2.1 Type I. Type I toilet soap shall be cake form. The cakes shall be firm, homogeneous, smooth, distinct, and uniform in shape and size. Each soap cake shall be formed by the pressing and stamping method.

3.2.1.1 Type I, class 1 cakes shall be manufactured to float on water.

3.2.1.2 Type I, class 2 cakes shall be milled.

3.2.2 Type II. Type II toilet soap shall be a clean, uniform, homogeneous, free-flowing powder.

3.3 Color.

3.3.1 Type I, class 1. The color of a freshly cut surface of a floating soap cake shall not be darker than color No. 37778 of Fed. Std. No. 595.

3.3.2 Type I, class 2. Unless otherwise specified in the procurement documents, milled soap cakes shall either be uncolored with the color of a freshly cut surface of a soap cake not darker than color No. 37778 of Fed. Std. No. 595, or tinted in pastel shades of pink, blue, green, yellow, or tan. If the soap is dyed, the hands shall not be stained by the use of the product. The manufacturer shall certify that the dyes used are certified by the Food and Drug Administration.

3.3.3 Type II. Type II toilet soap shall be uncolored or tinted in pastel shades of pink, green, or tan. If the soap is dyed, the hands shall not be stained by use of the product. The manufacturer shall certify that the dyes used are certified by the Food and Drug Administration.

3.4 Odor. Toilet soap shall be mildly perfumed and shall have a mild, pleasant odor.

3.5 Lather. Type I toilet soaps shall lather freely when tested as specified in 4.3.2.

3.6 Performance in dispenser. Type II toilet soap shall dispense freely, without caking, from a dispenser conforming to type III of FF-D-396, when tested as specified in 4.3.3.

3.7 Accelerated aging. The soap shall not become rancid or darken greatly when tested as specified in 4.3.5.

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3.8 Cake weight. The cake weight shall be uniform and within the tolerance of the nominal cake weight specified (see 6.2).

3.9 Workmanship. The finished product shall be clean and homogeneous in appearance and texture.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Certificates of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Quality conformance inspection.

4.2.1 Inspection of the end item. Components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, or qualified in this specification or applicable purchase documents.

4.2.2 Examination of preparation for delivery. Packaging, packing, and marking shall be examined to determine compliance with the requirements of section 5. Defects shall be scored in accordance with table II. Sampling and acceptance shall be in accordance with MIL-STD-105, inspection level S-2, acceptable quality level (AQL) of 4.0 percent defective. The sample unit shall be one shipping container fully prepared for delivery. The lot shall consist of all shipping containers presented for delivery at one time.

TABLE II. Examination of preparation for delivery

Examine	Defect
Container	Not as specified.
Contents	Not as specified.
Markings	Omitted; incorrect; illegible, improper size, location, sequence, or method of application.
Materials	Component missing or damaged.
Workmanship	Any nonconforming component; components missing, damaged, or otherwise defective.

4.2.2.1 Examination of closure, waterproofing, and banding of containers. When shipping containers are required to comply with PPP-B-636, examination for defects in closure, waterproofing, and banding shall be made in accordance with the appendix of that specification.

4.2.3 Lot. A lot for the purposes of examination and testing of the end item shall be as defined in 4.2.3.1 or 4.2.3.2.

4.2.3.1 Batch soap. For soap made by a batch process, a lot shall consist of all toilet soap of the same type and class which was made in the same batch and offered for delivery at one time.

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4.2.3.2 Continuous process soap. For soap made by a continuous process, a lot shall consist of all toilet soap of the same type and class which has been manufactured during one twenty-four hour period or part thereof, from the same raw material lots, under the same processing conditions and which is offered for delivery at one time.

4.2.4 Examination of the end item. The end item shall be examined for defects in accordance with table III. Sampling and acceptance shall be in accordance with MIL-STD-105, inspection level I, AQL of 6.5 percent defective. The sample unit shall be one soap cake for type I toilet soap and one unit package (see 5.1) for type II toilet soap. The lot shall be as defined in 4.2.3.

TABLE III. Examination of the end item

Examine	Defect
Type I:	
Odor	Objectionable, rancid, not mildly perfumed.
Cakes	Not as specified; dirty; presence of foreign matter; not pressed and stamped; not milled (class 2).
Wrapping	Not as specified.
Workmanship	Cakes not homogeneous in appearance and texture.
Color	Not as specified. Not uniform.
Type II:	
Unit package (see 5.1.1.2 or 5.1.2.2)	Not as specified.
Contents:	
Net weight	Less than specified.
Odor	Rancid, objectionable; not as specified.
Color	Not as specified.
Workmanship	Any sifting of contents; Contents not clean and homogeneous.

4.2.4.1 Examination of the end item for net weight (type I). The end item shall be examined for net weight. The sample unit shall be one cake of soap. The lot shall be as defined in 4.2.3. Sampling shall be in accordance with MIL-STD-105, inspection level I. Weigh each cake selected, and determine the average cake weight. Failure of the average cake weight to comply with the requirement of 3.8 shall be cause for rejection of the lot.

4.2.5 Testing of the end item.

4.2.5.1 Type I. Type I toilet soap shall be tested as specified in table IV. The sample unit shall be three cakes of soap. The lot shall be as defined in 4.2.3. Sampling shall be in accordance with MIL-STD-105, inspection level S-1. Two cakes from each sample unit shall be shaved, grated or shredded, and thoroughly mixed to form a composite sample. Place the composite sample in a clean, dry, watertight container. Seal the container and mark with the lot number and date. Perform each test twice. The lot shall be rejected if any sample fails any of the specified tests.

4.2.5.2 Type II. Type II toilet soap shall be tested as specified in table IV. The sample unit shall be one unit package (see 5.1). The lot shall be as defined in 4.2.3. Sampling shall be in accordance with MIL-STD-105, inspection level S-1. Take a one pound portion of toilet soap from each unit package so chosen and mix them thoroughly to form a composite sample. Place the composite sample in a clean, dry, watertight container and seal the container. Mark the container with the lot number and the date. Perform each test twice. The lot shall be rejected if the composite sample fails any of the specified tests.

TABLE IV. Instructions for testing the end item

Characteristic	Specification Reference		Results reported as:	
	Requirement	Test Method	Pass or Fail	Average, Numerically to nearest
Moisture and volatile matter	Table I	ASTM D 460 #11		0.1 percent
Alcohol insoluble matter	Table I	ASTM D 460 #16, 17		0.1 percent
Free alkali (as NaOH)	Table I	ASTM D 460 #18		0.1 percent
Matter insoluble in water	Table I	ASTM D 460 #19		0.1 percent
Anhydrous soap	Table I	ASTM D 460 #21, 22		0.1 percent
Unsaponified saponifiable matter	Table I	ASTM D 460 #36		0.1 percent

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TABLE IV. Instructions for testing the end item (con.)

Characteristic	Specification Reference: Requirement	Specification Reference: Test Method	Results reported as: Pass or Fail	Results reported as: Average, Numerically to nearest
Rosin	Table I	4.3.6	X	
Titer	Table I	ASTM D 460 #43, 44		0.1 degree C
Acid number	Table I	ASTM D 460 #45, 46		1.0 unit
Sodium chloride	Table I	ASTM D 460 #50-52		0.1 percent
Sugar	Table I	Fed. Std. 536 # 1701	X	
Fineness	Table I	Fed. Std. 536 # 2101		0.1 percent
Floating test	3.2.1.1	4.3.1		0.01 unit
Lather	3.5	4.3.2	X	
Performance in dispenser	3.6	4.3.3	X	
Staining of the hands	3.3.2, 3.3.3	4.3.4	X	
Accelerated aging	3.7	4.3.5	X	

4.3 Test methods.

4.3.1 Floating test. Fill a container to a depth of at least three inches with warm water (100-115 degrees F). When placed in the water, the soap cake shall float with the top surface of the cake above the water.

4.3.2 Lather. Pre-wet both hand palms in warm water (100° - 115°F), having a hardness of not less than 5 nor more then 10 grains per gallon, calculated as $CaCO_3$. Give a fresh soap cake 30 brisk strokes between the palms of both hands. The amount of lather produced shall be enough to partially obscure the palms from view.

4.3.3 Performance in a dispenser. Fill a soap dispenser conforming to type III of FF-D-396 with 21.8 cubic inches (12 fluid ounces) of the composite sample. Operate the feed valve of the dispenser with full, uniform strokes until the composite sample is entirely removed. It shall take no less than 800 strokes and no more than 2,600 strokes to empty the dispenser. The composite sample shall not clog or render inoperative any vital operating parts of the dispenser.

4.3.4 Staining of the hands. Wash the hands with a soap cake or 1 gram of the composite sample and warm tap water. Rinse the hands well and dry with a clean, white towel. The hands shall not be stained by the soap.

4.3.5 Accelerated aging. Place 10 grams (g) of the composite sample in each of two clean glass bottles. Place one bottle in an oven maintained at 65° ± 1°C and allow sample to reach oven temperature. Tightly cap the bottle with a cap that is not harmed by the oven temperature and keep sample at oven temperature for 96 hours. Tightly cap the other bottle and keep at room temperature. Remove the sample from the oven and allow to cool to room temperature. After the sample has cooled, remove the cap and examine the sample. The sample shall not have a rancid odor, nor have darkened greatly when compared with the sample kept at room temperature.

4.3.6 Qualitative test for rosin (Liebermann-Storch). Transfer 1 to 2 milliliters of the fatty acids prepared in section 42 of ASTM Standard D 460 to a test tube, treat with 5 to 10 milliliters of reagent grade acetic anhydride, and warm on a steam bath. After cooling, pour 1 to 2 milliliters into a white porcelain dish and allow a drop or two of sulfuric acid (sp. gr. 1.53) to run down the side of the vessel. If rosin is present, a fugitive violet coloration changing to a brownish tinge is immediately produced at the margin of contact of the reagents. This test can be made on a smaller sample on a spot plate. The tests should be checked with a sample of pure stearic acid to which about 3 percent of rosin has been added. (Sulfuric acid of 1.53 specific gravity is prepared by diluting 34.7 ml. of sulfuric acid (sp. gr. 1.84) with 35.7 ml. of distilled water.)

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5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, B, or C, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Type I. Unless otherwise specified (see 6.2), soap cakes shall be furnished unwrapped. When wrapped cakes are specified, the wrapping shall be of a type normally used by the industry.

5.1.1.2 Type II. Five pounds of powdered toilet soap shall be packaged in a box conforming to PPP-B-566, variety 2, style 1, type A, class "b". The box shall be completely sealed with a permanent type of adhesive applied to the full area between the top and bottom flaps to prevent any sifting of the contents.

5.1.2 Level B.

5.1.2.1 Type I. Unless otherwise specified (see 6.2), soap cakes shall be furnished unwrapped. When wrapped cakes are specified, the wrapping shall be of a type normally used by the industry.

5.1.2.2 Type II. Five pounds of powdered toilet soap shall be packaged in a paperboard box conforming to PPP-B-566, variety 1, style 1, type A, class "b". The paperboard shall be double kraft lined chip calandered to a finish 2 of L-P-378. The paperboard shall have a minimum of 0.032 caliper and basis weight of 62 pounds. The box shall be completely sealed with an adhesive applied to the full area of the flaps to prevent any sifting of contents.

5.1.3 Level C. Toilet soap shall be packaged to afford adequate protection against loss or damage during shipment from the supply source to the first receiving agency.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Type I. Cakes of one class and weight, in the quantity specified in table V, shall be packed in a box conforming to PPP-B-636, class weather resistant. If the cakes are unwrapped, they shall have a sheet of water resistant paper or plastic separating the layers and protecting the upper and lower surfaces. The box shall be provided with a liner conforming to MIL-L-10547, type I, except that the liner shall be fabricated of a 0.002 inch (2 mil) thick polyethylene film or a laminate of 30-pound basis weight (24 x 36 - 500) kraft paper and 0.001 inch thick polyethylene film. The bottom and side heat-scaled seams shall be not less than 3/8 inch in width. When closed, the case liner shall be folded flat against the contents. The closure shall be a heat-seal or a lockseam. For the heat-sealed closure, excess air shall be expelled prior to sealing. The box shall be closed, waterproofed, and banded in accordance with the appendix of PPP-B-636.

5.2.1.2 Type II. Ten 5-pound boxes of powdered toilet soap shall be packed in a box conforming to PPP-B-636, class weather resistant. The box shall be closed, waterproofed, and banded in accordance with the appendix to PPP-B-636.

TABLE V. Number of cakes per shipping container

Nominal cake weight	Number of cakes per shipping container
3/4 ounce avoirdupois (oz av) ± 1/4 oz av	1000
2 oz av ± 1/2 oz av	500
3 oz av ± 1/2 oz av	144
6 oz av ± 1/2 oz av	100

5.2.2 Level B.

5.2.2.1 Type I. Cakes of soap of one class and weight, in quantities as specified in table V, shall be packed in a box conforming to PPP-B-636, class domestic. If the cakes are unwrapped, they shall have a sheet of water-resistant paper or plastic separating the layers and protecting the upper and lower surfaces. The box shall be closed in accordance with the appendix of PPP-B-636.

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5.2.2.2 Type II. Ten 5-pound boxes of powdered toilet soap shall be packed in a box conforming to PPP-B-636, class domestic. The box shall be closed in accordance with the appendix of PPP-B-636.

5.2.3 Level C. Toilet soap, in quantities as specified (see 6.2), shall be packed in containers to assure carrier acceptance and safe arrival at destination. Containers shall comply with the National Motor Freight Classification or Uniform Freight Classification, as applicable.

5.3 Marking. In addition to any special marking required by the procurement documents, interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123 for civil agencies, or MIL-STD-129 for military agencies, as applicable (see 6.2).

6. NOTES

6.1 Intended use. Type I cake soaps are intended primarily for use with soft water to clean the skin. Type I, class 1 floating soap may also be used for hand laundering, dishwashing, and general cleaning purposes where a mild soap is required. Type II powdered soap is intended to be dispensed from a dispenser conforming to FF-D-396, type III, and used with soft water for hand cleaning.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type and class required (see 1.2).
(c) Nominal cake weight required (see table V).
(d) Cake color required (if applicable, see 3.3.2).
(e) Whether cakes are to be individually wrapped.
(f) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
(g) Marking document required (see 5.3).

6.3 Supersession data. This document supersedes P-S-00620C (GSA-FSS), dated August 10, 1970; P-S-620B, dated August 24, 1965; and type I of P-S-00625D (GSA-FSS), dated October 29, 1970.

6.4 Classification change. Type I under P-S-00620C (GSA-FSS) is now type I, class 1 under this document. Type II under P-S-00620C (GSA-FSS) is now type I, class 2 under this document. Type I under P-S-00625D (GSA-FSS) is now type II under this document.

Former classification	New classification
P-S-00620C (GSA-FSS), type I	P-S-620D, type I, class 1
P-S-00620C (GSA-FSS), type II	P-S-620D, type I, class 2
P-S-00625D (GSA-FSS), type I	P-S-620D, type II

6.5 Bid evaluation. Bids should be evaluated on the basis of price per pound of soap.

6.6 Payment. Payment should be made on the basis of the adjusted net weight of soap received. To determine the amount to be paid to the contractor, multiply the price per pound of soap in the contract by the adjusted net weight of soap received as calculated in 6.6.1 or 6.6.2.

6.6.1 Adjusted net weight of type I, class 1 floating cakes.

$$\text{Adjusted net weight} = \frac{R \times (100-M)}{80}$$

6.6.2 Adjusted net weight of type I, class 2 milled cakes.

$$\text{Adjusted net weight} = \frac{R \times (100-M)}{88}$$

Where R = actual net weight of soap received.
M = percent moisture and volatile matter at 105°C in the soap as received (see 4.2.5).

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6.7 The Department of Defense has waived coordination of this document until further notice.

MILITARY INTEREST:

User Activity:

Army - GL

CIVIL AGENCY COORDINATING ACTIVITY:

GSA - FSS

PREPARING ACTIVITY: GSA - FSS

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